



NATIONAL JUNIOR COLLEGE
SH 1 PROMOTIONAL EXAMINATIONS

Higher 2

CANDIDATE
NAME

SUBJECT
CLASS

REGISTRATION
NUMBER

CHEMISTRY

9729/02

Paper 2 Structured Questions and Free Response Questions

Tues 1 Oct 2019

Additional Materials: Data Booklet
Writing Paper

2 hours 30 min

READ THE INSTRUCTIONS FIRST		For Examiner's Use	
Write your name, subject class and registration number on all the work you hand in. Write in dark blue or black ink on both sides of the paper. You may use a soft pencil for any diagrams or graphs. Do not use paper clips, highlighters, glue or correction fluid. Section A: Structured Questions Answer ALL questions on the question paper. Section B: Answer ALL questions from this section in the Answer Booklet. The use of an approved calculator is expected where appropriate. A Data Booklet is provided. The number of marks is given in brackets [] at the end of each question or part question. Appropriate significant figures and units are expected for final numerical answers. Submit answers to Section A and Section B separately.		Section A:	
		1	/4
		2	/12
		3	/17
		4	/11
		5	/15
		Section B:	
		1	/22
		2	/19
		Penalty	
		s.f.	
		units	
		Total	/100

This document consists of **18** printed pages.

Section A: Structured Questions [Total: 59 marks]Answer **ALL** questions in the space provided.

- 1 (a) A sample of lead contains four stable isotopes with the following percentage abundances.

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Isotope	Percentage abundance / %
^{204}Pb	1.4
^{206}Pb	24.1
^{207}Pb	22.1
^{208}Pb	a

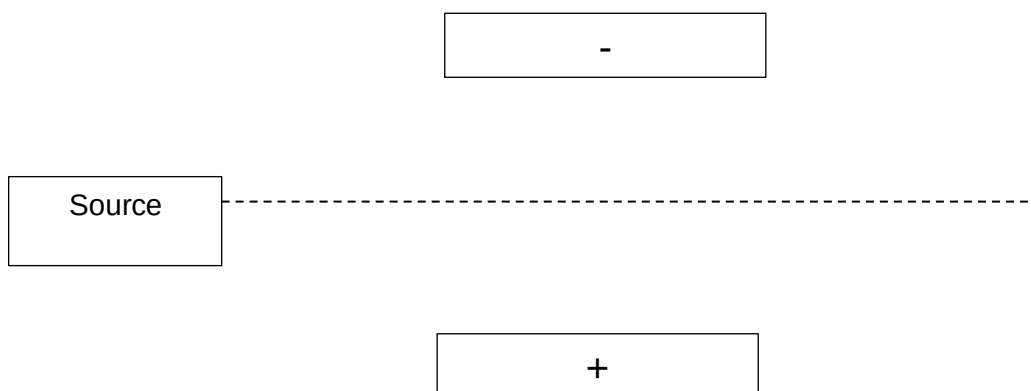
Determine the value of **a**.

Hence calculate the relative atomic mass of lead in the sample. Give your answer to two decimal places.

[2]

- (b) Beams of charged particles are deflected by an electric field. The particles are travelling at the same speed through an electric field of constant strength.
In a particular experiment, protons are deflected through an angle of $+26^\circ$.

A beam of $^{24}\text{Mg}^{2+}$ ions was subjected to an identical set of experimental conditions.
Calculate the angle deflected, giving your answer to 2 decimal places.

Draw the paths taken by both the protons and $^{24}\text{Mg}^{2+}$ ions in the diagram below.

[2]

[Total: 4]

- 2 (a) (i) Butane is used as a propellant in hair spray to create an aerosol mist of liquid particles. The bottle of hair spray has an initial temperature of 27°C . When the hair spray is used up, only butane gas remained. The internal pressure in the bottle decreased to 138 kPa.

Calculate the new internal pressure when the temperature of the bottle reaches 50°C on a hot afternoon.

[1]

- (ii) As a safety precaution, the propellant has to be completely expelled from the bottle prior to incineration. Explain qualitatively why this precaution is necessary.

.....
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[1]

- (b) One mean of measuring toxicity is using LD, which stands for "Lethal Dose". LD₅₀ is the amount of a material which causes the death of 50% of a group of test animals. LD₅₀ value is expressed as the mass of a chemical administered per kg body mass of a test animal.

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Another means of measuring toxicity is using LC, which stands for "Lethal Concentration". The concentration of the chemical in air that kills 50% of the test animals during the observation period is the LC₅₀ value.

The table below shows the values for the LD₅₀ and LC₅₀ along with the toxicity ratings. (1 g = 1000 mg)

Toxicity rating	Commonly used term	LD ₅₀ : Oral (mg kg ⁻¹)	LC ₅₀ : Inhalation (ppm)
1	Extremely Toxic	1 or less	10 or less
2	Highly Toxic	>1 – 50	11 – 100
3	Moderately Toxic	51 – 500	101 – 1000
4	Slightly Toxic	501 – 5000	1001 – 10,000
5	Practically Non-toxic	5001 – 15,000	10,001 – 100,000

- (i) 4.45×10^{-4} mol of a toxic compound, C₄H₅NO, was found to cause death in 50 % of test animals weighing 1 kg.
Calculate LD₅₀ of the compound and state its toxicity rating.

[2]

- (ii) Phosphine gas, PH₃, is widely used in the semi-conductor industry as a dopant. The concentration of a small quantity of gas is usually expressed in parts per million (ppm) as shown below:

$$\text{Concentration in ppm} = \frac{\text{volume of gas}}{\text{volume of air}} \times 10^6$$

Given that LC₅₀ for PH₃(g) is 200 mg m⁻³ at room temperature and pressure, convert LC₅₀ to ppm and determine its toxicity rating.

[2]

- (c) (i) The name of the element, chromium, means colour in Greek because many chromium compounds are intensely coloured.

State the full electronic configuration of a chromium atom in its ground state.

.....[1]

- (ii) Draw and label the shapes of all the d orbitals, showing the axes clearly.

[2]

- (d) The successive ionisation energy of any element can be determined in the laboratory.

- (i) Write an equation for the fifth ionisation energy for sulfur.

.....[1]

- (ii) Explain why the fifth ionisation energy of sulfur and chlorine is different.

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[2]

[Total:12]

- 3** Hematite is a common iron oxide with the formula Fe_2O_3 . It is a very important naturally occurring compound that finds widespread use as a heterogeneous catalyst.

(a) (i) Describe how a heterogeneous catalyst speeds up gaseous reactions.

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[3]

(ii) Using relevant values from the *Data Booklet* and information provided in the table below, draw a Born-Haber cycle and use it to calculate the lattice energy of Fe_2O_3 .

Enthalpy change of atomisation of iron	/ kJ mol^{-1}	+ 415
Enthalpy change of formation of Fe_2O_3	/ kJ mol^{-1}	– 765
First electron affinity of oxygen	/ kJ mol^{-1}	– 141
Second electron affinity of oxygen	/ kJ mol^{-1}	+ 790

[4]

- (iii) From the knowledge of the distances between the cations and anions in the crystal structures and the charge on each ion, it is also possible to calculate theoretical values for lattice energies.

Given that the theoretical value of lattice energy of Fe_2O_3 is $-14774 \text{ kJ mol}^{-1}$, account for the discrepancy between your answer in **a(ii)** and the theoretical value.

.....

[2]

- (iv) Fe_2O_3 can also react with CO to produce Fe.



Calculate $\Delta H_f^\ominus$ for $\text{CO}(\text{g})$, given that $\Delta H_f^\ominus$ for $\text{CO}_2(\text{g}) = -394 \text{ kJ mol}^{-1}$ and using suitable values from the table in **(a)(ii)**.

[2]

- (b) Fe^{2+} ions are also commonly used as a homogenous catalyst in reactions involving two anions.

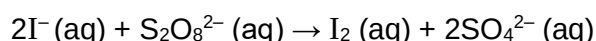
- (i) The reaction between peroxodisulfate ions, $\text{S}_2\text{O}_8^{2-}$ and iodide ions is very slow at room condition.

Explain why this is so.

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[1]

- (ii) Write two equations to show how Fe^{2+} catalyse the reaction below.



Equation 1:

Equation 2:

[2]

- (iii) With the aid of a Boltzman distribution curve, explain how Fe^{2+} ions catalyse the reaction between $\text{S}_2\text{O}_8^{2-}$ and iodide ions.

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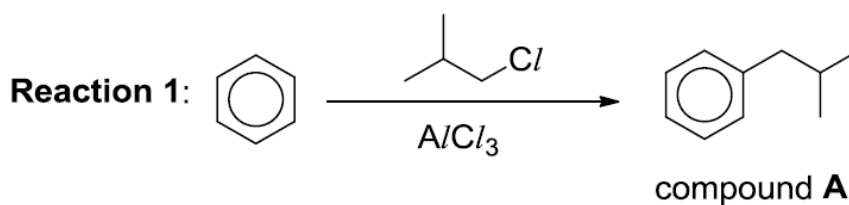
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[3]

[Total: 17]

- 4 | Compound **A** can be synthesised from benzene and 1-chloro-2-methylpropane in the presence of aluminium chloride as shown in Reaction 1.



- (a) | Describe the mechanism for Reaction 1. You should show all relevant charges, lone pairs, dipoles and movement of electrons by curly arrows.

[3]

- (b) Compound **B**, which contains a chiral centre and is an isomer of **A**, is also formed in Reaction 1. **B** is formed after the electrophile undergoes rearrangement through the movement of an alkyl group to an adjacent carbon bearing the positive charge.

(i) Give the structure of **B**.

[1]

(ii) Suggest the structure of the electrophile that gives rise to **B** and explain why the formation of this electrophile is favoured.

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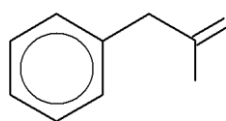
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[3]

(c) Compound **C** has the structure as shown below.



Suggest a simple chemical test to distinguish between compounds **A** and **C**.

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[2]

- (d) Suggest whether **C** would be soluble in water.

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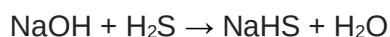
[2]

[Total: 11]

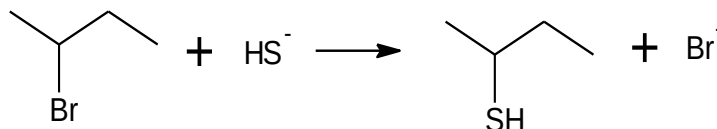
- 5 Many thiols have strong odors resembling that of garlic or rotten eggs. Thiols are used as odorants to assist in the detection of natural gas.

The general formula of thiols is R-SH where R is an alkyl group.

Thiols are formed when halogenoalkanes react with hydrogen sulfide ions via nucleophilic substitution. Hydrogen sulfide ions are generated as shown below:



One such example is the reaction of 2-bromobutane with hydrogen sulfide ion via $\text{S}_{\text{N}}1$ mechanism.



- (a) Describe the mechanism of the above reaction, showing relevant lone pairs and dipoles, and use curly arrows to indicate the movement of electron pairs.

[2]

- (b) Explain why the resultant product is not optically active.

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[2]

- (c) What do you understand by the term *nucleophile*? Explain why hydrogen sulfide ion is a stronger nucleophile than hydrogen sulfide.

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[2]

- (d) The kinetics of the reaction between 1-bromobutane and hydrogen sulfide ions, HS^- , was studied and the following results were obtained.

Initial concentration of 1-bromobutane /mol dm ⁻³	Initial concentration of HS^- /mol dm ⁻³	Initial rate of reaction /mol dm ⁻³ s ⁻¹
0.1	0.1	1.5×10^{-5}
0.2	0.1	3.0×10^{-5}
0.3	0.2	9.0×10^{-5}
x	y	4.5×10^{-5}

- (i) Deduce the orders of reaction with respect to 1-bromobutane and HS^- ions.

[2]

- (ii) Hence write the rate equation and determine the rate constant of the reaction.

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[2]

- (iii) Suggest a set of values for x and y .

[1]

- (iv) Using the rate equation in d(ii), suggest whether the reaction between 1-bromobutane and hydrogen sulfide ions would follow the same mechanism described in (a).

.....
.....

[1]

- (e) Two other thiols, **Y** and **Z**, have the following percentage composition by mass: C, 71%; H, 7.9%; S, 21.1%. Each molecule of **Y** and **Z** contains only 1 sulfur atom. Only **Z** is optically active and decolourises hot acidified potassium manganate(VII).

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[Note: S-H is inert to hot acidified potassium manganate(VII).]

- (i) Determine the molecular formula of **Z**, and suggest its structure.

[2]

- (ii) Suggest the structure of **Y**.

[1]

[Total:15]

Section B: Free Response Questions [Total: 41 marks]

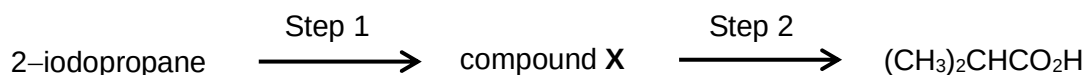
Answer both questions in the Answer Booklet.

- 1 The elements in Group 17, the halogens, and their compounds, show many similarities and trends in their properties. The table below shows some data for the elements, fluorine to iodine.

Element	Bond energy / kJ mol^{-1}	Standard enthalpy change of atomisation, $\Delta H^\circ_{\text{at}}$ / kJ mol^{-1}	Boiling point/ K	Boiling point of hydrogen halide / K
Fluorine, F–F	158	79	85	293
Chlorine, Cl–Cl	244	122	238	188
Bromine, Br–Br	193	112	332	206
Iodine, I–I	151	107	457	238

- (a) Explain the trend in the boiling points of the hydrogen halides, HF, HCl, HBr and HI. [3]
- (b) 0.60 g of a gaseous halogen occupies 100 cm^3 at 70°C and 1 atm.
- (i) Determine the relative molecular mass, M_r , of the halogen and deduce its identity. [2]
- (ii) Explain why the M_r calculated in **b(i)** is larger than expected. [1]
- (c) (i) The standard enthalpy change of formation of ICl(g) is $-24.0 \text{ kJ mol}^{-1}$. Use this information and the data in the table above to calculate the bond energy of I–Cl using a suitable energy cycle. [2]
- (ii) Propane can react with ICl to form 2-iodopropane. Describe the mechanism for the formation of 2-iodopropane, showing curly arrows. [3]
- (iii) Using relevant data from the *Data Booklet* and your answer in **c(i)**, verify that the formation of 2-chloropropane from the reaction described in **c(ii)** is always non-spontaneous. [3]

- (iv) 2-iodopropane can be converted to $(\text{CH}_3)_2\text{CHCO}_2\text{H}$ via a 2-step synthesis as shown.



Suggest the structure of compound **X** and give the reagents and conditions for each step.

[3]

- (d) Compound **O**, C_5H_{12} , is an alkane that reacts with chlorine gas to give only one monochlorinated product.

- (i) Give the structural formula of compound **O**.

[1]

Compound **P** is an isomer of **O**. Compound **P** reacts with chlorine to form several monochlorinated products but only one product **Q** contains a chiral centre. Under suitable conditions, **Q** forms only two isomeric products **R** and **S**, C_5H_{10} .

- (ii) Give the skeletal formulae of **P**, **Q**, **R** and **S**.

[4]

[Total: 22]

- 2 (a) Strontium chloride, SrCl_2 , is a typical salt which forms neutral aqueous solutions. It emits a bright red colour in a flame and this allows it to be used as a source of redness in fireworks.

Use the data below to answer the questions that follow.

	/kJ mol ⁻¹
Standard enthalpy change of solution of strontium chloride	-51.9
Standard enthalpy change of hydration of strontium ion	-1446
Standard enthalpy change of hydration of chloride ion	-378

- (i) Define the term *standard enthalpy change of solution* of strontium chloride. [1]

- (ii) Write an equation to show the relationship between the lattice energy and $\Delta H^\circ_{\text{sol}}$ of strontium chloride.

Hence, calculate the lattice energy of strontium chloride.

[2]

- (iii) Explain how the magnitude of the enthalpy change of hydration of bromide ions differs from that of chloride ions.

[2]

- (b) Mass spectrometry is an analytical technique which measures the masses of chemical species present in a sample. A mass spectrum can be used to determine the isotopic composition of a sample and to elucidate the chemical structure of a chemical compound.

The mass-to-charge ratio, also known as m/e value, is the M_r of chemical species present in the sample. The peak height ratio gives an indication of the isotopic ratio of an element in a sample.

In the first experiment, propene was reacted with liquid bromine in inert organic solvent. The organic products were then extracted and analysed using mass spectrometry. The mass spectrum is as shown in Figure 2.1.

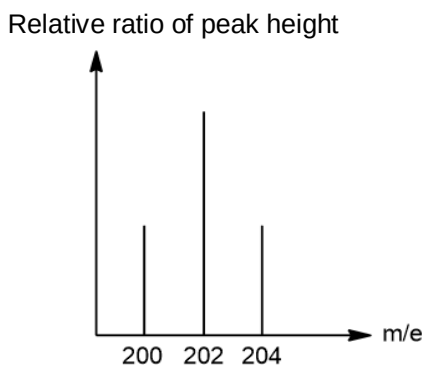


Figure 2.1. Mass spectrum of the products from reaction of propene with bromine

- (i) Draw the structure of the product formed when propene reacts with bromine in inert organic solvent.

[1]

- (ii) Based on the m/e values in Fig 2.1, deduce the two isotopes of bromine.

[1]

- (iii) Using your answers in **b(i)** and **b(ii)**, determine the relative abundance of the isotopes of bromine.

You may assume that carbon, hydrogen and oxygen only exist in one isotopic form of ^{12}C , ^1H and ^{16}O respectively.

[1]

- (iv) Describe the mechanism for the reaction between propene and bromine in inert organic solvent.
Show relevant lone pairs and dipoles, and use curly arrows to indicate the movement of electron pairs.

When propene was reacted with aqueous bromine, a different mass spectrum was obtained as shown in Fig 2.2. There are two additional peaks with greater peak height at m/e ratio of 138 and 140.

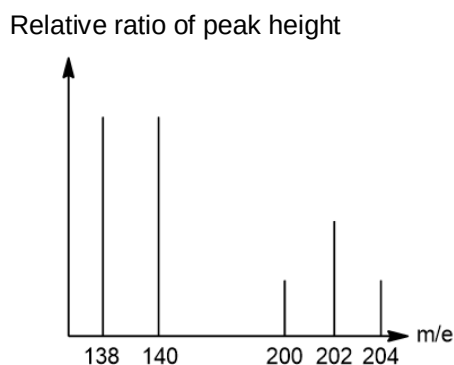


Fig 2.2 products from reaction of propene with aqueous bromine

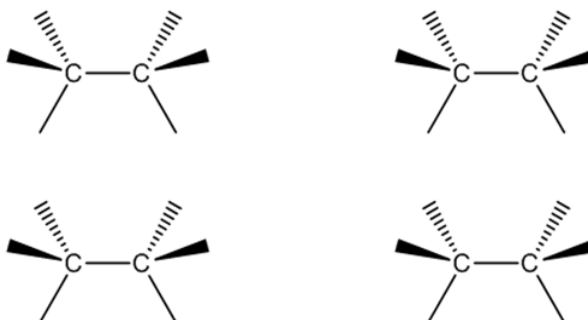
- (v) Draw a possible structure that accounts for the product with m/e value of 138.

[1]

- (vi) Explain, with reference to the mechanism you have described in **b(iv)**, the m/e and relative ratio of the two additional peaks.

[2]

- (vii) Draw the four different possible products that are responsible for the peak at m/e value of 138. Copy the diagram below to your answer booklet and complete the diagram.



[2]

- (viii) Ethanoic acid could be synthesised from propan-1-ol in two steps.

Suggest the structure of compound **Y** and give the reagents and conditions for each step.



[3]

[Total: 19]

End of Paper